OptiSched: Uncertainty-Aware Next-Term Enrollment Forecasting & CP-SAT Academic Scheduling

Bridging 10-Year Historical Registrar Data, Split Conformal Risk Bounds, and Constraint Programming for Conflict-Free University Timetables

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Team: Joe Doan, Tony Nguyen, Tina Nguyen, Salvatore Nigro
Faculty Advisor: Prof. Yugyung Lee

1. INNOVATION VS. EXISTING SCHEDULING SYSTEMS

Traditional university schedulers treat next-semester enrollment as a static number copied from last term, causing severe Week-1 overflows.

Capability	Existing Schedulers	OptiSched (Ours)
Demand Input	Prior-term copy or raw point estimate	**Next-term forecast + 80%/95% Conformal Bounds**
Historical Training	1 prior term	**10 years history (19 terms → next semester)**
Conflict & Prefs	Manual edits / greedy heuristics	**Exact OR-Tools CP-SAT + 3-Tier Faculty Prefs**
What-If Shocks	Manual spreadsheet trial	**Live Counterfactual Simulator + Bottleneck Proof**

2. END-TO-END SYSTEM ARCHITECTURE

1. Historical ETL
10-yr UIUC FOIA (22,991 sections → 800 course-terms; 19 train, 1 held-out).

2. Forecast + Bounds
Pooled Ridge next-term forecast + Split Conformal 80%/95% upper bounds.

3. CP-SAT Solver
Hard room/slot/load constraints + 3-tier faculty preferences (40/40/20).

4. Live What-Ifs
Enrollment surges, faculty buyouts, room closures & NL Copilot.

- **Important Clarification:** The model uses **10 years of historical data (Fall 2016 – Fall 2025)** to predict the **next semester (Spring 2026)**, rather than predicting 10 years into the future.
- **3-Tier Faculty Hierarchy:** Resolves per-course overrides (Tier 1), general semester availability (Tier 2), and 2-year causal teaching habits (Tier 3) across 40% Day, 40% Slot, and 20% Room weights.

3. VERIFIED HELD-OUT SPRING 2026 QUANTITATIVE RESULTS

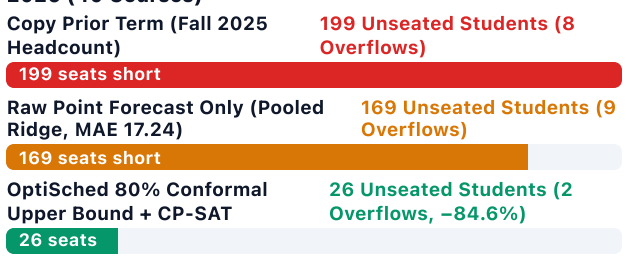
−84.6%
Unseated Students (169 → 26)

9 → 2
Overflowed Courses Cut

97.5% / 85%
95% (39/40) & 80% (34/40) Coverage

84.0%
2-Yr Faculty Habit Match (0 Conflicts)

Visual Comparison: Unseated Students on Held-Out Spring 2026 (40 Courses)



- **How This Benefits Students:** Prevents 143 Week-1 seat lockouts in core STEM courses, keeping students on track for prerequisites and on-time graduation.
- **How This Benefits Administrators:** Eliminates chaotic Week-1 room reassignments and emergency section splits while guaranteeing **0 room or instructor double-bookings (OPTIMAL)**.
- **Forecasting Accuracy:** On held-out Spring 2026, Pooled Ridge reaches **17.24 MAE** vs **18.03** for Seasonal Naive and **28.25** for copying prior term.

4. LIVE COUNTERFACTUAL WHAT-IF DEMONSTRATION

Operational Shock	What OptiSched Demonstrates Live
1. Enrollment Surge (CS +20%, STAT +15%)	Re-solves timetable with forecast sizing; upgrades surging classes into larger compatible halls with minimal ripple.
2. Instructor Unavailable (1-Course Buyout)	Proves **INFEASIBILITY** when no coverage is allowed (20×2=40 cliff); finds **1-course minimal move** with colleague/adjunct coverage.
3. Classroom Reassigned (0.9x Capacity)	Detects exact lab/auditorium capacity cliffs (e.g. ECE 380 exceeding 30-seat lab cap) and re-packs feasible rooms.

5. EVALUATION CLARIFICATION: REAL VS. SIMULATED

Component	Provenance & Scope
Real Registrar Data	800 course-semester enrollments across 40 STEM courses & 20 terms (Fall 2016 – Spring 2026) from UIUC FOIA records.
Simulated Scenarios	Campus classroom pool, 3-tier faculty preference weights, and What-If counterfactual shocks (surge, buyout, room closure).

6. EXTENSION: FACULTY COMMERCIALIZATION ADVISOR

As a connected extension for faculty whose research leads to a grant course buyout:

- **In-Memory UM-16C Disclosure Drafter:** Extracts 26 disclosure fields with verbatim quotes from uploaded research paper PDFs.
- **BM25 Policy Retrieval:** Searches **425 indexed passages across 7 UM System & UMKC policy documents** to generate a conflict-of-interest checklist.
- **Direct Scheduler Integration:** Triggers the live **CP-SAT Course-Release Simulator** to verify department teaching coverage before approving a buyout.

7. REPRODUCING THE DEMONSTRATION

Repo: `github.com/JoeDoan/CS5551_Advanced-Software-Engineering-`

1. Start Backend (FastAPI + OR-Tools CP-SAT):

```
cd backend && python3 -m uvicorn app.main:app --port 8000
```

2. Start Frontend (React + TypeScript + Vite):

```
cd frontend && npm run dev   # http://localhost:5173/insights
```

3. Run Automated Verification Suites (Backend + Frontend):

```
cd backend && pytest && cd ../frontend && npx vitest run
```